
LLMs are Bayesian in Expectation, Not Realization

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Abstract

Bayesian accounts of in-context learning appear to face a direct objection: exact posterior predictives for exchangeable data are invariant to task-preserving order, while transformers often change next-token probabilities when the same examples are serialized differently. We show that this objection targets a structural invariant rather than the operational quantity used to score online prediction. For any Bayesian predictive reference, excess sequential code length is exactly cumulative predictive KL. For unordered support sets that must be serialized, the expected regret of a single admissible ordering decomposes into the regret of the order-averaged predictor plus an order-averaging gain. Exchangeability violations are therefore not binary refutations; they are priced by log loss. We instantiate the theory with KT/Dirichlet finite-alphabet prediction, candidate-event decompositions, safe-code floors, and coarsened Bayesian linear-regression (BLR) predictive distributions. On Qwen2.5-7B/14B, floored candidate distributions at support 256 have one-step excess code lengths of 0.020/0.011 bits for Bernoulli prediction and 0.039/0.022 bits for four-way categorical prediction, with candidate mass above 0.999; coarsened BLR continuations increasingly match the posterior-predictive digit distribution as support grows. A frequentist plug-in baseline sharpens the reading: the predictive distributions sit closer to the Bayesian posterior predictive than to the maximum-likelihood plug-in, by a margin largest at small support—where the plug-in is degenerate—and vanishing as the references converge. Position interventions and a from-scratch encoding ablation separate task-preserving from semantic order and localize it to the positional encoding; activation patching tests whether decoded sufficient statistics causally affect predictions; and permutation mixtures quantify the log-loss cost of arbitrary orderings downstream. The result is a quantitative reconciliation: transformers need not realize exchangeable posterior predictives for every serialization to be Bayes-competitive sequential predictors.

1 Introduction

Bayesian interpretations of in-context learning give few-shot prompting a clean semantics: the support examples identify a latent task, and the continuation is the posterior predictive under that task [1, 2, 4, 3]. This view predicts a concrete object: the conditional distribution for the next token. It also appears to collide with a concrete fact about transformers. A Bayesian posterior predictive for exchangeable observations is invariant to task-preserving reordering; an autoregressive transformer reads a sequence with positions.

Recent martingale and exchangeability diagnostics make this collision explicit. Pretrained language models do not, in general, satisfy the identities of exact Bayesian prediction on exchangeable data:

permuting an unordered support set can change next-token probabilities, and posterior-predictive martingale tests can fail [9, 10]. These results are important. They rule out the strongest slogan version of Bayesian ICL: a transformer is not an exact exchangeable posterior-predictive distribution for every serialization of the same support set.

They do not rule out the operational claim that matters for online prediction. A sequence predictor is scored by prequential log loss,

$$L_n(Q, y_{1:n}) = \sum_{t=1}^n -\log q_t(y_t | y_{<t}), \quad (1)$$

where $q_t(\cdot | y_{<t})$ is the conditional distribution actually used at prefix t . This score does not directly penalize failure of invariance at unused serializations. It penalizes probability assigned to the next observation. If the model’s conditional distributions stay close to a Bayesian reference in predictive KL along the evaluated path, the cumulative code length can remain close to the Bayesian code even when exact exchangeability fails.

The central contribution is therefore a change of estimand. Structural exchangeability is a property of an exact posterior-predictive law. Prequential regret is the operational quantity paid by a sequence predictor. The bridge between them is predictive KL. We prove this first as an exact prequential comparison identity. We then show how arbitrary serialization of an unordered information state contributes an additional, measurable order-averaging gain. This turns exchangeability from a yes/no test into a code-length budget.

Thesis. Exchangeability failures refute exact posterior-predictive equivalence. They do not, by themselves, refute Bayes-competitive prequential prediction. The operational question is how much cumulative predictive KL the violations spend.

Contributions. First, we give a robust formalization of the evaluation target: excess prequential code length against any Bayesian reference is cumulative predictive KL. Second, we introduce a task-preserving ordering decomposition: the regret of a single serialization equals the regret of the order-averaged predictor plus a nonnegative order-averaging gain. Third, we connect language-model scoring to native probabilities through a candidate-event decomposition and safe-code floors with explicit overhead. Fourth, we audit pretrained transformers in closed-form Bayesian settings, coarsened BLR continuations, position interventions, causal statistic-use interventions, and downstream order-averaging experiments. Fifth, we add a non-Bayesian baseline—the maximum-likelihood and Laplace plug-in predictors—and report the model-to-Bayes minus model-to-plug-in predictive code length, which tests whether a frequentist sequential predictor accounts for the same conditionals. The experiments are not presented as evidence that transformers are literal posterior samplers. They estimate the code-length terms that determine whether order sensitivity is operationally costly.

Relation to prior work. Bayesian and latent-task accounts explain ICL as approximate posterior prediction [1, 2, 3]; algorithmic accounts show that transformers can implement learned predictors, gradient-descent-like updates, or algorithm selection in context [5, 6, 7, 4, 8]; positional-encoding work studies the sequence mechanisms that make exact exchangeability nontrivial [15, 16, 17, 18]; and prompt-order work documents practical sensitivity to support order and calibration [11, 12, 13, 14]. The martingale critiques make exact posterior-predictive equivalence untenable for present LLMs [9, 10]. We keep that negative result and replace the binary invariance question by a prequential regret question.

2 Prequential prediction under task-preserving orderings

We distinguish two notions that are often conflated. *Structural Bayesianity* asks whether every realized posterior-predictive distribution satisfies invariances implied by the probabilistic model, such as exchangeability under permutations of an unordered support set. *Operational Bayes-competitiveness* asks how much extra prequential code length a predictor pays relative to a Bayesian predictive reference. These notions are not equivalent.

2.1 Prequential comparison

Let P , B , and Q be sequential processes on a common dominated observation space, with conditional predictive distributions

$$p_t(\cdot | h_t), \quad b_t(\cdot | h_t), \quad q_t(\cdot | h_t), \quad h_t = y_{<t}.$$

Think of P as the evaluation distribution, B as the Bayesian reference, and Q as the model being scored.

Theorem 1 (Prequential comparison identity). *Assume the relevant absolute-continuity and integrability conditions. Then*

$$\mathbb{E}_P[L_n(Q) - L_n(B)] = \sum_{t=1}^n \mathbb{E}_{H_t \sim P} \mathbb{E}_{Y_t \sim p_t(\cdot | H_t)} \left[\log \frac{b_t(Y_t | H_t)}{q_t(Y_t | H_t)} \right]. \quad (2)$$

Equivalently,

$$\mathbb{E}_P[L_n(Q) - L_n(B)] = \text{KL}(P_{1:n} \| Q_{1:n}) - \text{KL}(P_{1:n} \| B_{1:n}). \quad (3)$$

In the special case $P = B$,

$$\mathbb{E}_B[L_n(Q) - L_n(B)] = \sum_{t=1}^n \mathbb{E}_{H_t \sim B} \text{KL}(b_t(\cdot | H_t) \| q_t(\cdot | H_t)). \quad (4)$$

The identity is elementary but decisive. Under a Bayesian reference process, excess code length is exactly cumulative predictive KL. Exchangeability violations matter operationally only through the predictive KL they accumulate. This identity is the paper’s formal replacement for binary martingale rejection as an evaluation target.

2.2 Serialization of unordered information states

An unordered statistical object must be serialized before a transformer can process it. Let Z_t denote the task-relevant information state at prefix t . A *task-preserving ordering* is a serialization that leaves Z_t unchanged. For Bernoulli prediction, $Z_t = (t, S_t)$; for categorical prediction, $Z_t = (t, c_1, \dots, c_K)$; for conjugate Bayesian linear regression (BLR), Z_t consists of the sufficient statistics and query covariate; for evidence-grounded QA, it is the evidence set under the admissible retrieval-order distribution. If reordering changes the label or the task semantics, it is not task-preserving and should not be averaged.

Let $\mu_t(\cdot | Z_t)$ be any distribution over task-preserving orderings. Uniform permutations are a special case; banded retrieval-preserving permutations are another. Let $q_t^g(\cdot | Z_t)$ be the model’s predictive distribution after serialization g , and define the order-averaged predictor

$$\bar{q}_t(\cdot | Z_t) = \mathbb{E}_{g \sim \mu_t(\cdot | Z_t)} q_t^g(\cdot | Z_t). \quad (5)$$

Theorem 2 (Order-averaging decomposition). *Suppose the Bayesian reference predictive distribution $b_t(\cdot | Z_t)$ depends only on Z_t and is therefore invariant to task-preserving orderings. Then, for each information state,*

$$\mathbb{E}_{g \sim \mu_t} \text{KL}(b_t(\cdot | Z_t) \| q_t^g(\cdot | Z_t)) = \text{KL}(b_t(\cdot | Z_t) \| \bar{q}_t(\cdot | Z_t)) + J_t(Z_t), \quad (6)$$

where

$$J_t(Z_t) = \mathbb{E}_{Y \sim b_t} [\log \bar{q}_t(Y | Z_t) - \mathbb{E}_{g \sim \mu_t} \log q_t^g(Y | Z_t)] \geq 0. \quad (7)$$

Consequently, the expected prequential regret of a sampled single ordering decomposes as

$$R_n^{\text{single}} = R_n^{\text{avg}} + \sum_{t=1}^n \mathbb{E}[J_t(Z_t)]. \quad (8)$$

Thus exact exchangeability corresponds to zero order-averaging gain. Nonzero gain is the extra log loss paid by using one admissible serialization instead of the averaged predictive distribution. The theorem does not make order sensitivity free. It prices it.

Corollary 1 (Persistent order sensitivity has a predictive price). *In natural-log units, on a shared finite candidate space,*

$$\mathbb{E}_{g, g'} \text{TV}(q_t^g, q_t^{g'})^2 \leq 2 \mathbb{E}_g \text{KL}(b_t \| q_t^g). \quad (9)$$

Therefore a predictor with small average predictive KL cannot have persistent large task-preserving order dispersion on the scored event space.

2.3 Candidate events and safe predictors

Language-model evaluations often score a fixed set of answer events rather than the entire vocabulary. Scoring a fixed answer set is valid only if the mass outside the event set is reported or decomposed. For single-token labels the events are tokens; for multi-token answers they are the continuation events defined by the chosen verbalizer. The following identity makes the decomposition exact.

Proposition 1 (Candidate-event decomposition). *Fix a prefix and mutually exclusive candidate events $E_1, \dots, E_m$, with $E = \cup_i E_i$. Suppose the reference distribution is supported on these events, $b_i = B(E_i | h)$ and $\sum_i b_i = 1$. Let the model assign event probabilities $Q(E_i | h)$ and total candidate mass*

$$M_Q(E | h) = \sum_i Q(E_i | h).$$

When $M_Q(E | h) > 0$, define the normalized candidate distribution $\bar{q}_i = Q(E_i | h)/M_Q(E | h)$. Then

$$\mathbb{E}_{i \sim b}[-\log Q(E_i | h) + \log b_i] = \text{KL}(b \| \bar{q}) - \log M_Q(E | h). \quad (10)$$

Candidate normalization therefore does not hide probability mass: it separates predictive-shape error from outside-candidate mass. We report both quantities in the controlled audits.

Lemma 1 (Safe-code floor). *Let q be a predictive distribution on m candidate events and let u be uniform. For $\lambda \in (0, 1)$ define*

$$q^\lambda = (1 - \lambda)q + \lambda u.$$

Then $q_i^\lambda \geq \lambda/m$ for all i . Moreover, whenever $\text{KL}(b \| q) < \infty$,

$$\text{KL}(b \| q^\lambda) \leq \text{KL}(b \| q) - \log(1 - \lambda), \quad (11)$$

and always

$$\text{KL}(b \| q^\lambda) \leq \text{KL}(b \| u) - \log \lambda \leq \log m - \log \lambda. \quad (12)$$

A fixed floor gives an explicit finite-horizon safe code; a vanishing floor schedule with $\sum_{t \leq n} -\log(1 - \lambda_t) = o(n)$ preserves asymptotic Bayes-competitiveness.

2.4 Closed-form instantiations

The theory is not Bernoulli-specific. Bernoulli and categorical tasks are the cleanest finite-alphabet settings because the Bayesian reference distribution is closed form.

Corollary 2 (Finite-alphabet KT/Dirichlet certificate). *Let $\mathcal{A}$ be finite and let K_{KT} be the Krichevsky–Trofimov (KT) Dirichlet($1/2, \dots, 1/2$) predictive code. For any i.i.d. source $p \in \Delta(\mathcal{A})$,*

$$\mathbb{E}_{p^n} L_n(K_{\text{KT}}) = nH(p) + O_{|\mathcal{A}|}(\log n). \quad (13)$$

For any predictor Q ,

$$\mathbb{E}_{p^n} L_n(Q) = \mathbb{E}_{p^n} L_n(K_{\text{KT}}) + \mathbb{E}_{p^n} \sum_{t=1}^n \log \frac{K_{\text{KT}}(Y_t | Y_{<t})}{q_t(Y_t | Y_{<t})}. \quad (14)$$

Thus if the cumulative comparison term is at most R_n , then Q has expected code length at most $nH(p) + O(\log n) + R_n$. The finite-support approximation and floor conditions reported in [Table 2](#) are one sufficient way to obtain $R_n = O(\sqrt{n})$.

Corollary 3 (Coarsened continuous predictive distributions). *Let B be a Bayesian predictive process with conditional density for a real-valued observation, and let V be a finite quantizer or verbalizer with cells $E_1, \dots, E_m$. The induced distributions*

$$b_t^V(i | h_t) = B(Y_t \in E_i | h_t), \quad q_t^V(i | h_t) = Q(Y_t \in E_i | h_t)$$

satisfy the same prequential comparison identity for the discrete process $V(Y_t)$. If full densities for B and Q exist, data processing gives $\text{KL}(b_t^V \| q_t^V) \leq \text{KL}(b_t \| q_t)$.

Table 1: Theory-to-evidence map. Each empirical quantity estimates a code-length term, a mass term, or a task-preservation condition in the theory.

Formal quantity	Interpretation	Main evidence
$\mathbb{E}_B[L_n(Q) - L_n(B)]$	Excess prequential code length	Table 2
$\text{KL}(b_t \ q_t)$	One-step predictive regret	Tables 2 and 3
$\text{KL}(b_t \ \bar{q}_t)$	Regret after averaging task-preserving orderings	Tables 9 and 15
J_t	Log-loss cost of using one admissible ordering	NLL/order-averaging gains; Section 4.4
$M_Q(E)$	Probability assigned to scored candidate events	candidate-mass column
$-\log(1 - \lambda)$	Safe-code overhead from flooring	native/floored comparison
Task-preserving Z_t	Whether reordering preserves the statistical task	Table 6 and section 4.4
Statistic intervention effect	Whether decoded statistics causally affect predictions	Table 8
$\text{KL}(b_t \ q_t) - \text{KL}(f_t \ q_t)$	Separation from a frequentist plug-in f_t (MLE/Laplace/OLS)	Tables 4 and 5

3 Experimental protocol

All experiments score next-token probabilities. The main models are Qwen2.5-7B/14B-Instruct and Qwen2.5-7B base for position and activation interventions; Llama-3.1-8B appears in probe and evidence-permutation comparisons [22, 23]. Apart from a controlled from-scratch ablation that isolates the positional-encoding channel ([Section 4.4](#)), all experiments are inference-only on pretrained checkpoints. Downstream order-averaging audits use MMLU, GSM8K, BBH, FEVER, HotpotQA, NQ-Open, and PopQA slices [31, 32, 33, 34, 35, 36, 37]. For discrete conjugate tasks, the prefix is an unordered sequence of symbols and the target is the next symbol. For BLR, the prefix is a raw numeric continuation task: $x_1, y_1 \ x_2, y_2 \ \cdots \ x_*,$ for one covariate, or $x_1, y_1, z_1 \ \cdots \ x_*, y_*,$ for two covariates. The BLR reference is the Gaussian posterior predictive distribution integrated over the digit bins used for token scoring.

Native, candidate, and floored distributions. Native probabilities measure model behavior. When a task specifies candidate events, we report the normalized candidate distribution and the candidate mass, as justified by [Proposition 1](#). For the finite-alphabet discrete audit we also score a floored safe predictor,

$$q_{\theta, \lambda}(a | h_t) = (1 - \lambda)K_{\theta}(a | h_t) + \lambda U_{\mathcal{A}}(a), \quad \lambda = 0.01, \quad (15)$$

where $U_{\mathcal{A}}$ is uniform over the candidate alphabet. The floored predictor is not used as evidence about native model probabilities. It is an evaluated safe code with explicit finite-horizon overhead under [Lemma 1](#).

Task-preserving versus semantic order. Order averaging is justified only for serializations that preserve the information state Z_t . The Bayesian prediction task presents an exchangeable support set, so demonstration order is task-preserving. The semantic-first control makes the first demonstration semantically relevant, so position is part of the task. This distinction is central: the intervention is not to average away every order effect, but to price the effects of arbitrary serialization when order is not part of the task.

4 Empirical evidence

The experiments follow the theory. We first audit closed-form Bayesian predictive distributions. We then ask whether order effects are task-preserving or semantic, whether decoded sufficient statistics are causally used, and how much log loss is saved by averaging admissible orderings. The controlled predictive-distribution audits are the main evidence for Bayes-competitiveness. The downstream mixture experiments are predictive-score diagnostics, not claims of large accuracy gains.

Table 2: Discrete Bayesian predictive-distribution audit. Candidate distributions use the floored safe predictor $q_{\theta,\lambda}$ with $\lambda = 0.01$. KL is in bits and equals one-step excess prequential code length under the Bayesian predictive distribution.

Family	Model	Support	Contexts	KL bits	ℓ_1	$\epsilon_{\infty,A}$	$\hat{C}_A$	$c_{\min,A}$	Cand. mass
Beta-Bernoulli, 0/1	7B	256	256	0.020	0.081	0.243	3.902	0.192	0.9998
Dirichlet-categorical, 0/1/2/3	7B	256	256	0.039	0.154	0.300	4.805	0.171	0.9996
Beta-Bernoulli, 0/1	14B	256	512	0.011	0.067	0.227	3.632	0.329	0.9998
Dirichlet-categorical, 0/1/2/3	14B	256	128	0.022	0.124	0.183	2.937	0.401	0.9994

Table 3: Coarsened BLR continuation distributions on Qwen2.5-7B. The model is scored on the next digit after raw pair or tuple sequences. The full Bayesian posterior predictive becomes a better distributional match as support grows; in the two-covariate task it beats the best misspecified control at support 100.

Task	Support	$\text{corr}(\mathbb{E}_\theta, \mathbb{E}_B)$	Dist. corr.	Best control $\mathbb{E}$ corr.	Best control dist. corr.	KL bits
Pair-next-number	25	0.935	0.808	n/a	n/a	0.388
Pair-next-number	50	0.974	0.886	n/a	n/a	0.238
Pair-next-number	100	0.988	0.934	n/a	n/a	0.131
Tuple-next-number	25	0.880	0.713	0.891	0.667	0.669
Tuple-next-number	50	0.935	0.789	0.902	0.713	0.439
Tuple-next-number	100	0.965	0.849	0.881	0.689	0.318

4.1 Controlled Bayesian predictive distributions

Table 2 reports the finite-alphabet quantities covered by Corollary 2. The Bernoulli and four-way categorical distributions are scored as floored KT/Dirichlet safe predictors. The KL column is one-step excess prequential code length in bits under the Bayesian reference distribution. The envelope constant $\hat{C}_A$ and the floor ratio $c_{\min,A}$ report finite-support sufficient conditions for the $O(\sqrt{n})$ certificate; candidate mass reports how much native probability lies on the scored events.

Table 3 audits coarsened continuous prediction. BLR is continuous, so we score the induced digit distribution rather than claiming to measure the full posterior density. This is exactly the setting of Corollary 3. The posterior-predictive distribution becomes a better match as support grows, and in the two-covariate task it beats the best misspecified control by support 100.

The finite discrete distributions give the cleanest certificate: they have small predictive KL, candidate mass above 0.999, and audited floor ratios well above the deterministic floor. The BLR distributions support the same operational interpretation on induced digit events: at support 100, Qwen2.5-7B tracks the one-covariate posterior-predictive expectation with correlation 0.988 and the two-covariate full BLR expectation with correlation 0.965. Additional 14B BLR predictive-regret sweeps are in Section C.5.

4.2 A frequentist plug-in does not explain the predictive distributions

The audits above report small predictive KL against a Bayesian reference, but a Bayesian reading is only forced if a non-Bayesian sequential predictor does not fit the same conditionals equally well. The natural null is the frequentist plug-in: predict the next symbol from the running maximum-likelihood estimate—the empirical frequency—or with Laplace add-one smoothing; for the continuous task it is the ordinary-least-squares (OLS) point estimate. We score these predictors on the same contexts and under the same $\lambda = 0.01$ floor as the model, and report the difference of predictive code lengths $\Delta = \text{KL}(\text{model} \parallel \text{Bayes}) - \text{KL}(\text{model} \parallel \text{plug-in})$, so that $\Delta < 0$ certifies the model’s conditional is closer to the Bayesian predictive than to the frequentist one (Tables 4 and 5).

Two properties separate the model from the plug-in. First, the maximum-likelihood plug-in is a *degenerate* code on short prefixes: it assigns zero probability to any symbol not yet observed and pays infinite prequential code length whenever such a symbol occurs. This is not a small-constant effect—it is every single-example categorical context and 75% of contexts at support 16—while the order-averaged predictor $\bar{q}$ reserves posterior mass for unseen symbols (up to 0.68 at support one) and holds its excess over the Bayesian reference below 0.2 bits throughout. The plug-in is inadmissible precisely where the model is admissible. Second, where the plug-in is non-degenerate the model

Table 4: Frequentist plug-in baseline for the discrete predictive audit (companion to Table 2). The order-averaged model predictive $\bar{q}$ is compared to the KT/Jeffreys Bayesian reference and to two frequentist plug-ins—the MLE (empirical frequency) and Laplace (add-one)—across a support sweep; all predictors carry the $\lambda=0.01$ safe-code floor (Lemma 1). $\Delta_{\text{MLE}} = \text{KL}(\bar{q} \parallel \text{Bayes}) - \text{KL}(\bar{q} \parallel \text{MLE})$ and likewise Δ_{Lap} ; $\Delta < 0$ means the model is closer to the Bayesian posterior predictive than to that plug-in. “MLE deg.” is the fraction of contexts on which the unfloored MLE assigns zero probability to a possible symbol (infinite code length). KL in bits. These use the order-averaged predictor $\bar{q}$ of Theorem 2, so the model–Bayes column lies below the single-ordering values in Table 2.

Family	Model	Supp.	KL($\bar{q} \text{B}$)	Δ_{MLE}	Δ_{Lap}	MLE deg.	Cand. mass
Beta–Bernoulli	Qwen2.5-7B	4	0.0111	−0.134	−0.017	52%	0.9071
		16	0.0152	−0.007	−0.004	25%	0.9858
		64	0.0094	−0.002	0.000	12%	0.9980
		256	0.0022	−0.000	0.000	8%	0.9996
	Qwen2.5-14B	4	0.0186	−0.147	−0.013	52%	0.9045
		16	0.0122	−0.007	−0.003	25%	0.9923
		64	0.0086	−0.002	0.001	12%	0.9988
		256	0.0010	−0.000	−0.000	8%	0.9998
Dirichlet-cat. (0/1/2/3)	Qwen2.5-7B	4	0.1159	−0.762	−0.012	98%	0.8630
		16	0.0504	−0.084	−0.004	75%	0.9690
		64	0.0150	−0.005	−0.002	53%	0.9964
		256	0.0049	−0.001	0.000	28%	0.9988
	Qwen2.5-14B	4	0.1316	−0.860	0.003	98%	0.8439
		16	0.0499	−0.083	−0.004	75%	0.9776
		64	0.0182	−0.005	−0.002	53%	0.9974
		256	0.0046	−0.000	−0.000	28%	0.9993

still sits on the Bayesian side: Δ_{MLE} is −3.3 bits (7B) and −3.8 bits (14B) for four-way categorical prediction at a single example, −0.76 bits at support four, and −0.001 bits by support 256, where the Bayesian and plug-in predictives provably coincide and no predictor is distinguishable. The separation from the *Laplace* plug-in is an order of magnitude smaller and of inconsistent sign; this is the expected signature rather than a weakness, because add-one smoothing is itself the Dirichlet(1) posterior predictive—the comparison is then against a member of the model’s own conjugate family, and $\bar{q}$ lands near the Jeffreys value rather than at the maximum-likelihood corner.

The coarsened BLR continuations show the same separation against the OLS plug-in (Table 5). The point estimate reproduces the Bayesian posterior *mean*, so the plug-in’s entire deficit is the query-dependent predictive-variance inflation $\sigma^2 + x_*^\top \Sigma_n x_*$ that a posterior carries and a point estimate discards. Accordingly $\Delta = \text{KL}(\text{Bayes} \parallel q) - \text{KL}(\text{OLS} \parallel q)$ is negative on 85–99% of contexts—by 0.65 bits at support four, shrinking with support—and the model’s per-query predictive variance correlates with the Bayesian one at 0.57–0.83. The pattern is identical for one and two covariates and replicates on Llama-3.1-8B. In every setting the separation concentrates in the small-data regime, where Bayesian and frequentist prediction genuinely diverge, and closes asymptotically—the behavior the prequential comparison identity (Theorem 1) predicts for a Bayes-competitive predictor, and the reason the support-256 cells alone cannot adjudicate the question.

4.3 Position interventions separate task-preserving order from semantic order

We hold Qwen2.5-7B’s weights and tokens fixed and change only `position_ids` at inference time. The normal condition uses monotone positions. The demo-local condition resets positions inside each demonstration block, preserving within-demonstration token order while removing the absolute demonstration slot. The random-offset condition preserves within-demonstration order but assigns each demonstration a random positional offset. We evaluate 64 items with 16 support permutations per item for each task and position mode and report 95% bootstrap confidence intervals.

This intervention estimates whether ordering is task-preserving or semantic. In the Bayesian prediction task, absolute demonstration position explains much of the serialization variation. In the semantic-first control, removing that channel damages performance because position carries task signal. A reviewer should therefore not read order sensitivity as one phenomenon: the same positional machinery can be task-preserving in one task and signal in another.

Table 5: Frequentist plug-in baseline for the BLR predictive audit (companion to Table 3). The model’s next-digit distribution is compared to the full Bayesian posterior predictive and to the OLS point-estimate plug-in (same homoscedastic noise, no posterior-variance inflation). $\Delta = \text{KL}(\text{Bayes}||q) - \text{KL}(\text{OLS}||q)$ in bits; “closer” is the fraction of contexts on which the model is nearer the full Bayesian predictive than the plug-in; “var. corr” is the correlation between the model’s and the Bayesian per-query predictive variance. The plug-in’s deficit is the missing query-dependent variance inflation; $\Delta < 0$ means the model is closer to Bayes.

BLR	Model	Supp.	KL(B q)	KL(OLS q)	Δ [95% CI]	closer	var. corr
1-cov. (pair)	Qwen2.5-7B	4	0.7744	1.4265	-0.652 [-0.749, -0.566]	95%	0.571
		8	0.5913	0.8854	-0.294 [-0.335, -0.252]	93%	0.618
		16	0.4726	0.6254	-0.153 [-0.183, -0.124]	86%	0.736
	Qwen2.5-14B	4	0.7453	1.3821	-0.637 [-0.715, -0.562]	95%	0.705
		8	0.5262	0.8080	-0.282 [-0.321, -0.244]	93%	0.750
		16	0.3852	0.5093	-0.124 [-0.147, -0.102]	87%	0.796
2-cov. (tuple)	Qwen2.5-7B	20	0.6230	0.8234	-0.200 [-0.215, -0.186]	95%	0.826
	Qwen2.5-14B	20	0.6777	0.8807	-0.203 [-0.217, -0.190]	95%	0.770

Table 6: Pretrained position intervention. Resetting demonstration-local positions reduces Bayesian prediction order variance by about $21\times$, while the semantic-order control degrades. Positive Δ NLL (negative log-likelihood) means higher loss than normal positions.

Task	Position mode	Order variance	NLL	Δ NLL vs. normal
Bayesian prediction	normal	$5.108 [4.480, 5.780] \times 10^{-3}$	0.6157 [0.6090, 0.6221]	n/a
Bayesian prediction	demo-local	$2.420 [2.136, 2.715] \times 10^{-4}$	0.6358 [0.6297, 0.6417]	+0.0201 [0.0168, 0.0232]
Bayesian prediction	random offsets	$5.343 [4.833, 5.855] \times 10^{-3}$	0.6831 [0.6792, 0.6868]	+0.0674 [0.0614, 0.0734]
Semantic first	normal	$2.174 [2.005, 2.360] \times 10^{-3}$	0.1236 [0.1205, 0.1269]	n/a
Semantic first	demo-local	$1.049 [0.959, 1.139] \times 10^{-2}$	0.6599 [0.6468, 0.6730]	+0.5363 [0.5229, 0.5491]
Semantic first	random offsets	$2.020 [1.816, 2.228] \times 10^{-2}$	0.3793 [0.3639, 0.3947]	+0.2557 [0.2407, 0.2707]

4.4 Controlled positional-encoding ablation

The pretrained intervention shows that position carries the order channel, but it cannot remove that channel from a fixed model. To test whether positional encoding is the *source* of task-preserving order sensitivity, we train small transformers from scratch on i.i.d. Bernoulli sequences and vary only the positional-encoding scheme: none, learned absolute, sinusoidal, RoPE, and ALiBi, with a pooled readout that is permutation-invariant when no encoding is supplied. Training and evaluation settings are in Table 16.

The schemes separate sharply. With no positional encoding, the within-prefix order variance is $\approx 3.7 \times 10^{-16}$, the exact-exchangeability endpoint to numerical precision: the predictive distribution does not move across serializations, so J_t is zero at every state. Learned absolute, sinusoidal, RoPE, and ALiBi instead sit at 10^{-8} to 10^{-6} , an eight-to-ten order-of-magnitude increase. Exchangeability failure is thus a property of the encoding a transformer is given, not of the architecture; Section A.8 gives one stylized channel by which a fixed encoding produces this dispersion.

These models also let us read the order-averaging decomposition against a closed-form reference. Scoring cumulative regret relative to the KT predictive code (Corollary 2) at $n = 200$, the permutation-averaged regret stays between 0.0017 and 0.0022 bits/token across all five schemes, while the worst sampled ordering costs 0.0049 to 0.0098 bits/token. This is Equation (8) read off a controlled system: R_n^{avg} is a few thousandths of a bit against the Bayesian code, and the single-ordering penalty $\sum_t \mathbb{E}[J_t]$ is the worst-case-minus-average gap. Order sensitivity is real and bounded away from zero, and it is also small in the units that score prediction.

4.5 Decoded statistics are available and causally used

High probe accuracy alone would not establish Bayesian prediction; it only shows that a statistic is available [24, 25]. We therefore use probes as a localization diagnostic and activation patching as the causal-use test [26, 27, 28]. Ridge probes on residual streams recover the joint sufficient statistic (S_t, t) with $R^2 = 0.9998$ on Qwen2.5-7B and Llama-3.1-8B; categorical joint counts reach

Table 7: Linear-probe availability diagnostic. Count and position information is linearly available; Table 8 tests whether the information is used in the next-token distribution.

Model	Joint (S_t, t) R^2	Emp. freq. R^2	Pseudo-count R^2	Cat. joint R^2
Qwen2.5-7B	0.9998	0.904	0.926	0.9993
Llama-3.1-8B	0.9998	0.779	0.801	0.9992

Table 8: Layer-specific activation-patching audit on Qwen2.5-7B. Values are restoration scores with 95% bootstrap CIs. Learned subspace patching restores task logits far above matched random controls.

Task	Layer	Full vector	Learned subspace	Random subspace	Random vector
Keybinding	24	0.510 [0.503, 0.518]	0.254 [0.237, 0.270]	0.008 [0.006, 0.011]	-0.001 [-0.004, 0.001]
Keybinding	26	0.756 [0.744, 0.769]	0.338 [0.315, 0.361]	0.015 [0.011, 0.018]	0.006 [0.002, 0.009]
Bernoulli-next	24	0.887 [0.838, 0.939]	0.880 [0.814, 0.946]	-0.047 [-0.090, -0.005]	-0.014 [-0.064, 0.034]
Bernoulli-next	26	0.928 [0.887, 0.969]	0.854 [0.799, 0.910]	-0.051 [-0.105, 0.003]	-0.027 [-0.079, 0.024]

$R^2 = 0.9993$ and 0.9992 respectively. Empirical-frequency and pseudo-count probes are weaker, consistent with the model carrying count and position features before readout.

The probe result motivates an intervention. We form clean/corrupt prompt pairs with different sufficient statistics, patch a learned low-rank activation subspace from clean into corrupt examples, and measure restoration of the candidate-logit difference. Matched random subspaces and matched random vectors control for intervention norm and rank.

The learned subspace restores the candidate logit difference well above matched random controls, especially for Bernoulli-next. This closes the gap left by probes: beyond being decodable, the statistic can be intervened on to move the next-token distribution.

4.6 Order averaging as a predictive-score diagnostic

When support order is task-preserving, averaging predictive distributions across sampled serializations is the empirical version of Theorem 2. Concavity of log loss supplies the guaranteed comparison to the average sampled ordering; the magnitude of the gain estimates how much log loss arbitrary serialization spends. On MMLU, the effect is statistically precise but practically modest. That is consistent with the theory: the measured order-averaging gain is small for this benchmark and prompt distribution.

The mean per-item standard deviation of the gold-answer probability across random orderings is 0.0158 (95% CI [0.0151, 0.0165]), so order changes the predictive distribution even when it usually does not change the argmax. Among 359 mixture-wrong items, 319 are wrong under every sampled order and only 40 have any sampled order with the gold answer on top. Uniform averaging smooths the predictive distribution; it does not search for a favorable order. GSM8K-derived multiple choice shows the same calibration geometry at a larger scale: averaging random demonstration orders improves NLL from 1.5424 at $k = 1$ to 1.4948 at $k = 8$, Brier from 0.8052 to 0.7922, and expected calibration error (ECE) from 0.3204 to 0.3097, while accuracy remains near 0.43 (Table 14).

4.7 Evidence-grounded QA with exchangeable evidence

A practical version of the same problem arises when unordered evidence chunks must be serialized in a prompt. On a five-benchmark factuality slice of 3,059 items with 3–60 chunks per item, we form banded evidence permutations rather than all $n!$ permutations, reflecting a non-uniform admissible ordering distribution. Qwen2.5-7B-Instruct shows a dispersion slope of 0.377 versus $\log n$ and an order-averaging gain of 0.1041 nats/token; Llama-3.1-8B-Instruct shows a slope of 0.147 and a gain of 0.00982 nats/token. Uniform weights are within the measured optimization tolerance in both cases (Table 15). This is the applied order-averaging principle: if the evidence set is exchangeable but the model must read a sequence, score or average admissible serializations rather than treating one arbitrary order as the task.

Table 9: MMLU random-permutation audit on Qwen2.5-7B. With 1200 items, 8 demonstrations, and 16 unique random permutations per item, the same-sample predictive mixture improves NLL over the mean sampled single ordering. CIs are paired bootstraps; p -values are sign-flip tests.

Comparison	Baseline	Mixture ($k=16$)	Paired delta	p
Canonical accuracy	0.6983 [0.6717, 0.7242]	0.7008 [0.6750, 0.7275]	+0.0025 [−0.0042, 0.0100]	0.654
Canonical NLL	0.7243 [0.6705, 0.7768]	0.7212 [0.6694, 0.7741]	−0.0031 [−0.0074, 0.0009]	0.144
Mean single-order accuracy	0.6987 [0.6728, 0.7231]	0.7008 [0.6750, 0.7275]	+0.0021 [−0.0024, 0.0069]	0.378
Mean single-order NLL	0.7235 [0.6717, 0.7762]	0.7212 [0.6694, 0.7741]	−0.00230 [−0.00262, −0.00201]	0.00020

5 Boundary conditions and interpretation

The theory is broad, but each application must specify the predictive reference, the candidate events being scored, and the orderings that preserve the task. Bernoulli and categorical prediction use closed-form KT/Dirichlet references. BLR uses the induced digit distribution of a continuous posterior predictive, not the full continuous density. MMLU, GSM8K, and evidence QA are downstream order-averaging diagnostics unless an explicit Bayesian reference is defined.

The results do not show that transformers instantiate literal posterior distributions, nor that all order effects are harmless. [Corollary 1](#) states the opposite: persistent large movement of the scored predictive distribution has a log-loss price. The point is that the price is quantitative. Exact exchangeability is sufficient for zero order-averaging gain, but Bayes-competitive prequential prediction only requires the accumulated predictive KL to be small.

The floored discrete predictor is an evaluated safe code over the candidate alphabet and should be distinguished from native model behavior. Fixed flooring is a finite-horizon device with explicit overhead; asymptotic claims require native lower bounds or a vanishing floor schedule. Candidate normalization is interpreted through [Proposition 1](#), which is why candidate mass is reported. Finally, order averaging is appropriate only when order is task-preserving. For language modeling, chronology, chain-of-thought trajectories, or tasks where the first example changes the label, averaging orderings changes the task.

Broader impacts. The work is primarily diagnostic. A positive impact is more reliable evaluation of in-context predictors: order sensitivity can be priced by log loss rather than treated as a binary failure. A possible negative impact is that order averaging may improve calibration of systems used for automated adjudication without addressing factual, fairness, or deployment risks. We therefore frame the evidence-QA experiment as a predictive-score audit, not as a deployment recommendation.

6 Conclusion

The apparent paradox is resolved by separating structural invariance from operational regret. Transformers are not, in general, exact exchangeable posterior-predictive machines. But online prediction is scored by prequential log loss, and the excess over a Bayesian reference is cumulative predictive KL. When an unordered support set must be serialized, the additional cost of using one admissible ordering is the order-averaging gain. The experiments estimate these quantities: controlled discrete tasks give small KT/Dirichlet predictive KL, BLR continuations track coarsened posterior predictives, position interventions and a from-scratch encoding ablation separate arbitrary serialization from signal and bound its worst-case cost against the KT code, activation patching tests causal statistic use, and downstream mixtures measure serialization cost. A frequentist plug-in baseline separates the predictive distributions from a non-Bayesian sequential predictor, most sharply in the small-data regime and vanishingly as the references converge. Exchangeability failures therefore refute exact posterior equivalence, not Bayes-competitive prequential prediction.

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A Proofs and auxiliary formal statements

A.1 Proof of the prequential comparison identity

For any realized sequence,

$$L_n(Q) - L_n(B) = \sum_{t=1}^n \log \frac{b_t(Y_t | Y_{<t})}{q_t(Y_t | Y_{<t})}.$$

Taking expectation under P and conditioning on $H_t = Y_{<t}$ gives Equation (2). For the KL form, use the chain rule

$$\log \frac{P_{1:n}(Y_{1:n})}{Q_{1:n}(Y_{1:n})} = \sum_{t=1}^n \log \frac{p_t(Y_t | H_t)}{q_t(Y_t | H_t)},$$

and the analogous identity with B in place of Q . Subtracting expectations under P gives Equation (3). If $P = B$, then $p_t = b_t$, and the conditional expectation at each prefix is exactly $\text{KL}(b_t(\cdot | H_t) \| q_t(\cdot | H_t))$.

A.2 Proof of the order-averaging decomposition

Fix Z_t and suppress it from notation. Since $\bar{q}(y) = \mathbb{E}_g q^g(y)$,

$$\mathbb{E}_g \text{KL}(b \| q^g) = \mathbb{E}_g \sum_y b(y) \log \frac{b(y)}{q^g(y)} = \sum_y b(y) \log \frac{b(y)}{\bar{q}(y)} + \sum_y b(y) [\log \bar{q}(y) - \mathbb{E}_g \log q^g(y)].$$

The first term is $\text{KL}(b \| \bar{q})$ and the second is $J_t(Z_t)$. Non-negativity follows from Jensen’s inequality, because $\log \mathbb{E}_g q^g(y) \geq \mathbb{E}_g \log q^g(y)$ for each y . Summing the identity over t and taking expectation over Z_t gives Equation (8).

A.3 Proof of the dispersion-price corollary

Pinsker’s inequality in natural-log units gives $\text{TV}(b, q^g)^2 \leq \frac{1}{2} \text{KL}(b \| q^g)$. By the triangle inequality,

$$\text{TV}(q^g, q^{g'}) \leq \text{TV}(q^g, b) + \text{TV}(b, q^{g'}),$$

so $(a + b)^2 \leq 2a^2 + 2b^2$ gives

$$\text{TV}(q^g, q^{g'})^2 \leq 2 \text{TV}(q^g, b)^2 + 2 \text{TV}(q^{g'}, b)^2.$$

Taking expectation over independent g, g' and applying Pinsker proves

$$\mathbb{E}_{g, g'} \text{TV}(q^g, q^{g'})^2 \leq 4 \mathbb{E}_g \text{TV}(q^g, b)^2 \leq 2 \mathbb{E}_g \text{KL}(b \| q^g).$$

A.4 Proof of the candidate-event decomposition

Because $Q(E_i | h) = M_Q(E | h) \bar{q}_i$,

$$\mathbb{E}_{i \sim b} [-\log Q(E_i | h) + \log b_i] = \sum_i b_i \log \frac{b_i}{M_Q(E | h) \bar{q}_i} = \text{KL}(b \| \bar{q}) - \log M_Q(E | h).$$

A.5 Proof of the safe-code floor lemma

The lower bound $q_i^\lambda \geq \lambda/m$ is immediate. Since $q_i^\lambda \geq (1 - \lambda)q_i$,

$$\text{KL}(b\|q^\lambda) = \sum_i b_i \log \frac{b_i}{q_i^\lambda} \leq \sum_i b_i \log \frac{b_i}{(1 - \lambda)q_i} = \text{KL}(b\|q) - \log(1 - \lambda).$$

Also $q_i^\lambda \geq \lambda u_i$ gives

$$\text{KL}(b\|q^\lambda) \leq \text{KL}(b\|u) - \log \lambda \leq \log m - \log \lambda.$$

Summing the first overhead bound over t gives the stated condition for a vanishing floor schedule.

A.6 Proof of the finite-alphabet KT certificate

For a finite alphabet, the KT/Dirichlet($1/2, \dots, 1/2$) sequential code satisfies the standard redundancy bound

$$\mathbb{E}_{p^n} L_n(K_{\text{KT}}) = nH(p) + O_{|\mathcal{A}|}(\log n)$$

for any i.i.d. source p [19, 21, 20]. For any predictor Q ,

$$L_n(Q) - L_n(K_{\text{KT}}) = \sum_{t=1}^n \log \frac{K_{\text{KT}}(Y_t | Y_{<t})}{q_t(Y_t | Y_{<t})}.$$

Taking expectation gives Equation (14). If the comparison term is bounded above by R_n , the code-length bound follows.

One sufficient finite-support condition. Let $q_t^B = K_{\text{KT}}(\cdot | Y_{<t})$ and suppose a predictor q_t satisfies on the evaluated support

$$\|q_t - q_t^B\|_\infty \leq C(t+1)^{-1/2}, \quad q_t(a) \geq cq_t^B(a) \quad \forall a \in \mathcal{A}.$$

The same argument as in the KT redundancy proof bounds the expected one-step replacement cost by $O_{|\mathcal{A}|, C, c}((t+1)^{-1/2})$: for the realized symbol,

$$\log \frac{q_t^B(Y_t)}{q_t(Y_t)} \leq \frac{|q_t(Y_t) - q_t^B(Y_t)|}{q_t(Y_t)} \leq \frac{C}{c\sqrt{t+1}} \frac{1}{q_t^B(Y_t)},$$

and the Dirichlet-1/2 predictive denominator gives a bounded expectation of $p_a/q_t^B(a)$ after summing over symbols. Summing over t yields $R_n = O(\sqrt{n})$ for fixed alphabet and constants. This is the finite-support certificate reported in Table 2.

A.7 Proof of the coarsened-predictive corollary

The induced variables $V(Z_t)$ form a discrete sequential process with predictive distributions b_t^V and q_t^V , so Theorem 1 applies directly. If full densities exist, the partition map V is a measurable coarsening. Data processing for KL gives $\text{KL}(b_t^V\|q_t^V) \leq \text{KL}(b_t\|q_t)$.

A.8 Stylized positional mechanism

The main theorem does not assume a transformer mechanism for order sensitivity. The following calculation, adapted from a Bernoulli positional model, illustrates one sufficient source. Fix a length- t Bernoulli prefix with S_t ones and draw a uniform ordering τ . Suppose the next-token logit depends on the ordering through

$$u_{t,\tau} = \frac{1}{t} \sum_{i=1}^t a_{\tau,i} \text{PE}(i)$$

and is L_t -Lipschitz in this summary. Let

$$\sigma_{\text{PE},t}^2 = \frac{1}{t} \sum_{i=1}^t \left\| \text{PE}(i) - \frac{1}{t} \sum_{j=1}^t \text{PE}(j) \right\|^2.$$

Then the standard deviation, over within-prefix orderings preserving (t, S_t) , of the expected log score is bounded by

$$L_t \sigma_{\text{PE},t} \sqrt{\frac{S_t(t - S_t)}{t^2(t - 1)}} \leq \frac{L_t \sigma_{\text{PE},t}}{2\sqrt{t - 1}}.$$

This is not used as a transformer theorem in the main text. It only shows how positional encodings can create task-preserving order dispersion while the prequential theory above prices whatever dispersion is actually observed. Both constants are observable on white-box models: $\sigma_{\text{PE},t}$ is computed directly from the encoding and the context length, and L_t can be estimated from the gradient norm of the candidate logit with respect to $u_{t,\tau}$, so on the open-weight models audited here the bound is a measurable quantity rather than a free constant. The $t^{-1/2}$ factor predicts that within-prefix order dispersion shrinks as the prefix grows; this is the controlled-microscope counterpart of the finite-support order variances reported for the pretrained model in Table 6 and for the trained-from-scratch models in Section 4.4.

B Mechanistic sketch for implicit pseudo-counts

Self-attention can implement the implicit pseudo-count picture in Table 7 by pooling token embeddings in a way that depends mainly on the multiset of observed symbols. If queries and keys separate $x_i = 1$ from $x_i = 0$, the head output approximates a function of S_t (or $\hat{p}_t = S_t/t$). Given features that track (t, S_t) , an MLP readout can approximate the Beta–Bernoulli predictive mean for some effective pseudo-counts (α_0, β_0) :

$$\mathbb{P}(X_{t+1} = 1 \mid X_{1:t}) = \frac{\alpha_0 + S_t}{\alpha_0 + \beta_0 + t}.$$

This sketch explains why the predictive-distribution audit target is plausible in a controlled architecture. The causal intervention in Section 4.5 tests whether a pretrained transformer uses the statistic.

C Additional diagnostic and semantic-order results

C.1 Prefix-difference coefficients

The probes in Table 7 define availability and prefix-difference diagnostics on a real model. The relevant object is a prefix-state update. Let $r_\ell(p)$ denote the chosen readout residual at layer ℓ for prefix p . Let G be the matrix whose entry $G_{c,k}$ is the projected prefix finite difference, conditioned on appending the observed symbol c :

$$G_{c,k} = \mathbb{E}[(r_\ell(p_{t-1} \| z_t) - r_\ell(p_{t-1}))^\top w_{\ell,k} \mid z_t = c]. \quad (16)$$

An additive count diagnostic predicts $G \approx I$ for Dirichlet-categorical, slope-1 for Poisson, and $G \approx \mathbf{1}$ for Bernoulli. Define the additivity coefficient $\alpha_{\text{add}} := \overline{\text{diag } G} - \max_{i \neq j} |G_{ij}|$, and the concentration drift $\beta_{\text{conc}} := \text{median}(\alpha_{\text{eff}}/\alpha_{\text{true}})$ where α_{eff} is the effective prior concentration that minimises $\text{KL}(q_B \| q_M)$. Operationally, the reported probes instantiate r_ℓ at a fixed readout position and compare the distribution reached before and after appending the new observation. On Qwen2.5-7B Dirichlet ($K = 3$) at residual layer $\ell = 3$ the diagonal of G is $[1.016, 0.986, 1.002]$ with $\max_{i \neq j} |G_{ij}| = 0.020$, giving $\alpha_{\text{add}} = 0.981$; on Qwen2.5-7B Poisson–Gamma with $\alpha_{\text{true}} = 2.0$ the per-sequence median fitted prior is $\alpha_{\text{eff}} = 3.5$ ($\beta_{\text{conc}} = 1.75$, with mean-correlation 0.923 between model and reference predictives indicating concentration drift with preserved direction). These coefficients are planning diagnostics: they indicate when count summaries or concentration errors are the next object to test.

C.2 Activation-patching audit details

Table 8 reports Qwen2.5-7B activation patching with layer-specific learned subspaces. For each clean/corrupt pair, we compute the clean and corrupt candidate logit difference, filter pairs with small denominators, and score restoration as

$$\frac{\text{LD}_{\text{patch}} - \text{LD}_{\text{corrupt}}}{\text{LD}_{\text{clean}} - \text{LD}_{\text{corrupt}}}.$$

Subspaces are fit on separate calibration pairs at each patched layer. The matched random-subspace control samples a random basis orthogonalised against the learned subspace and rescales its component to the learned component norm; the matched random-vector control uses a random vector with the same norm. Keybinding uses 256 evaluation pairs. Bernoulli-next generates 256 evaluation pairs and retains 229 after the denominator filter.

The threshold-question BLR audit is a negative control for interface choice. When the prompt asks whether y exceeds a threshold and scores Yes/No logits, the reference-threshold control does not align with the analytic survival-logit distribution even though final-layer hidden replacement reproduces source model distributions. This identifies the measurement as an interface mismatch. The sequence-completion BLR audits below use raw continuation distributions and track posterior-predictive expectations directly.

C.3 BLR pair-next-number audit

The BLR pair audit samples one-dimensional integer covariates $x \in \{0, \dots, 9\}$, generates noisy linear responses, rounds and clips y to digits $0, \dots, 9$, and presents the support as a raw sequence

$$x_1, y_1 \ x_2, y_2 \ \cdots \ x_m, y_m \ x_\star,$$

with no instruction and no chat template. The model distribution is the normalized next-token distribution over digits $0, \dots, 9$. The reference is the Gaussian BLR posterior predictive distribution integrated over the same digit bins. Table 10 reports the support sweep.

Support	Contexts	$\text{corr}(\mathbb{E}_\theta[y], \mathbb{E}_B[y])$	Dist. corr.	Top-1 match	$\mathbb{E}[y]$ MAE
25	256	0.935	0.808	0.480	0.459
50	256	0.974	0.886	0.590	0.299
100	512	0.988	0.934	0.676	0.207

Table 10: **BLR pair-next-number audit.** The model is scored on the next digit after a raw pair sequence. Expectation and distributional correlations increase with support.

C.4 Multivariate BLR tuple-next-number audit

The multivariate BLR audit uses the same next-token setup with two digit covariates. Each context samples $x, y \in \{0, \dots, 9\}$, generates a noisy linear response from a model with design vector $[1, x, y]$ after digit normalization, rounds and clips z to digits $0, \dots, 9$, and presents the support as

$$x_1, y_1, z_1 \ x_2, y_2, z_2 \ \cdots \ x_m, y_m, z_m \ x_\star, y_\star,$$

again with no instruction and no chat template. The model distribution is the normalized next-token distribution over z digits. The main reference is the Gaussian BLR posterior predictive distribution for the full $[1, x, y]$ design, integrated over the same digit bins. Misspecified posterior controls use x-only, y-only, and intercept-only designs. Table 11 reports the support sweep.

Support	Contexts	$\text{corr}(\mathbb{E}_\theta[z], \mathbb{E}_B[z])$	Dist. corr.	Best control $\mathbb{E}[z]$ corr.	Best control dist. corr.	Top-1 match
25	256	0.880	0.713	0.891	0.667	0.465
50	256	0.935	0.789	0.902	0.713	0.535
100	512	0.965	0.849	0.881	0.689	0.557

Table 11: **Multivariate BLR tuple-next-number audit.** The model’s next- z digit distribution is compared with the full $[1, x, y]$ posterior predictive and the best misspecified control. At support 100, the full two-covariate reference is the best match.

C.5 BLR predictive-regret support sweep

This sweep reports raw full-vocabulary predictive regret for the same pair and tuple continuations used in Tables 10 and 11. The KL column is $\text{KL}(K_B \| K_\theta)$ in bits. The ℓ_1 and $\epsilon_{\infty, \mathcal{A}}$ columns use raw candidate probabilities, with candidate-set leakage included in ℓ_1 . These distributions support the BLR reference-comparison claim; the floored discrete KT distribution is reported in Table 2.

Model	Family	Support	Contexts	KL bits	ℓ_1	$\epsilon_{\infty, \mathcal{A}}$	$c_{\min, \mathcal{A}}$
7B	BLR pair-next-number	25	256	0.388	0.538	0.539	0.0571
7B	BLR pair-next-number	50	256	0.238	0.398	0.453	0.0296
7B	BLR pair-next-number	100	256	0.131	0.288	0.320	0.0092
7B	BLR tuple-next-number	25	256	0.669	0.719	0.775	0.0183
7B	BLR tuple-next-number	50	256	0.439	0.549	0.801	0.0069
7B	BLR tuple-next-number	100	256	0.318	0.462	0.659	0.0151
14B	BLR pair-next-number	25	256	0.309	0.468	0.459	0.0357
14B	BLR pair-next-number	50	256	0.251	0.398	0.516	0.0107
14B	BLR pair-next-number	100	256	0.160	0.303	0.385	0.0102
14B	BLR tuple-next-number	25	256	0.761	0.769	0.769	0.0218
14B	BLR tuple-next-number	50	256	0.456	0.556	0.772	0.0143
14B	BLR tuple-next-number	100	256	0.325	0.470	0.594	0.0145

Table 12: **BLR predictive-regret support sweep.** One-covariate and two-covariate BLR continuations are scored by raw next-token probability against the posterior predictive digit distribution. Increasing support reduces KL regret for both models and both BLR families.

C.6 Semantic-order and calibration observations

Table 13 applies the same averaging operation to semantic-order settings. These cells measure semantic-distribution and trajectory-style variance reduction, separated from the exchangeability experiments in the main text.

Model	Task	Acc $k=1$	Acc $k=16$	NLL $k=1$	NLL $k=16$
Qwen2.5-7B-Instruct	BBH-navigate	0.650	0.644	1.862	1.468
Qwen2.5-7B-Instruct	BBH-logical-deduction	0.730	n/a	0.836	n/a
Qwen2.5-7B-Instruct	BBH-tracking	0.260	0.266	2.628	2.471
Qwen2.5-7B-Instruct	GSM8K-CoT-reorder	0.990	1.000	0.052	0.014

Table 13: **Semantic-order controls.** These cells separate semantic-distribution and trajectory-style variance reduction from the exchangeable, task-preserving ordering experiments.

Calibration and accuracy are tracked separately from predictive-mixture NLL. In the random-permutation 1200-item MMLU audit, canonical accuracy is 0.6983 and mixture accuracy is 0.7008, while the same-sample NLL comparison improves from 0.7235 to 0.7212. In the GSM8K-derived multiple-choice run, NLL, Brier, and ECE improve monotonically from $k = 1$ to $k = 8$ while accuracy stays near 0.43. In distributional terms, this is a margin/entropy transformation of $K_\theta(\cdot | p)$: the top token can remain stable while probability mass and logit gaps move under averaging. This is consistent with prior calibration observations for instruction-tuned models [11, 12, 29, 30].

D Full task-preserving ordering mixture tables

k	Accuracy	NLL	Brier	ECE
1	0.4324 ± 0.0033	1.5424 ± 0.0056	0.8052 ± 0.0023	0.3204 ± 0.0023
2	0.4263 ± 0.0028	1.5161 ± 0.0050	0.7983 ± 0.0019	0.3159 ± 0.0029
4	0.4295 ± 0.0026	1.5018 ± 0.0026	0.7940 ± 0.0012	0.3096 ± 0.0028
8	0.4292 ± 0.0017	1.4948 ± 0.0018	0.7922 ± 0.0008	0.3097 ± 0.0014

Table 14: **GSM8K-derived few-shot multiple choice.** Averaging random demonstration orderings improves NLL, Brier, and ECE while leaving accuracy nearly flat, consistent with order-averaged predictive scoring.

	Qwen2.5-7B-Instruct	Llama-3.1-8B-Instruct
Dispersion slope b (vs. $\log n$)	0.377 ([0.319, 0.435])	0.147 ([0.109, 0.184])
Order-averaging gain (nats/token)	0.1041	0.00982
Mixture optimality gap (nats/token)	$< 10^{-4}$	$\leq 5.3 \times 10^{-5}$

Table 15: Evidence-grounded QA under evidence permutations. Both instruction-tuned models show order dispersion and positive order-averaging gains from uniform permutation mixtures. Uniform weights are within the measured optimization tolerance.

E Reproducibility

Table 16 summarizes the experimental configuration. All experiments use bfloat16 inference on a single A100 (40 GB) or H100 (80 GB) GPU per cell, with a fixed prompt-sampling seed (SEED=0). For reproducibility, a complete supplementary package should include anonymized code, configurations, prompt templates, seeds, and per-cell JSON outputs; any de-anonymized release should be deferred until camera-ready. Dataset manifests for MMLU, BBH, GSM8K are used under their published licenses; FEVER, HotpotQA, NQ-Open, and PopQA are used under CC-BY-SA 4.0 / CC-BY 4.0 / MIT respectively.

F Statistical analysis

Bootstrap confidence intervals use the percentile method, resampling independent units: items for the pretrained position intervention and MMLU audit, paired examples for activation patching, sequences for Bernoulli gap experiments, and tasks for ICL k -sweeps. MMLU paired deltas use sign-flip p -values over item-level deltas. Model comparison on gap-vs- n uses weighted least squares with inverse-variance weights $w_n = N_n / \hat{\Delta}_n^2$.

G Computational resources

The reported open-weight experiments use bfloat16 or 4-bit single-GPU inference on A100 40 GB, H100 80 GB, or equivalent 7B–14B workers. The position intervention and causal-patching cells require model-internal access; scoring cells require saved next-token log probabilities on the audited candidate sets. Resampling is performed over saved per-item or per-example outputs, so bootstrap and sign-flip intervals do not require model re-execution.

Component	Configuration
PE intervention (Table 6)	Qwen2.5-7B with fixed pretrained weights; inference-only position_ids modes {normal, demo-local, random demo off-sets}; 64 items $\times$ 16 permutations per task and mode; support size 16; Bayesian prediction and semantic-first tasks; 2,000 bootstrap resamples over independent items.
From-scratch PE ablation (Section 4.4)	Small transformers trained on i.i.d. Bernoulli sequences, $d=128$, 4 layers, 4 heads, pooled [CLS] readout (permutation-invariant with no encoding); positional schemes {none, learned absolute, sinusoidal, RoPE, ALiBi}; AdamW, 2,000 steps, batch 128, learning rate 3×10^{-4} , prefix lengths sampled up to the context, Bernoulli rates from a Jeffreys Beta($\frac{1}{2}, \frac{1}{2}$) prior; within-prefix order variance from 500 permutations at $t \in \{20, 40\}$; KT regret at $n=200$ with 60 sequences and 20 permutations per sequence across $p \in \{0.05, 0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$.
Probes (Table 7)	scikit-learn Ridge($\alpha=1$); $N_{\text{seq}} \in \{150, 300\}$ prompts of length $T=100$; 75/25 train/val split; per-layer R^2 , best layer reported.
Bayesian and BLR distribution audits (Tables 2 and 12)	Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct; symbol, pair, and tuple continuations; raw full-vocabulary candidate-token probabilities and conditional candidate distributions saved; floored discrete distributions use $q_{\theta, \lambda}$ with $\lambda = 0.01$ candidate-uniform flooring; supports 1, 2, 4, 8, 16, 32, 64, 128, 256 for Bernoulli and categorical distributions and 25, 50, 100 for BLR distributions; reported metrics are $\text{KL}(K_B \ q_{\theta, \lambda})$ for floored discrete distributions, $\text{KL}(K_B \ K_{\theta})$ for BLR distributions, full-distribution $\ell_1, \epsilon_{\infty, A}, \hat{C}_A, c_{\min, A}$, candidate mass, and support-binned JSON outputs.
BLR pair-next-number (Table 10)	Qwen2.5-7B-Instruct, bfloat16/4-bit inference; raw prompts $x_1, y_1 \ x_2, y_2 \ \dots \ x_*$, with no instruction and no chat template; y candidates are digit tokens $0, \dots, 9$; supports 25, 50, 100 with 256, 256, 512 contexts respectively.
Multivariate BLR tuple-next-number (Table 11)	Qwen2.5-7B-Instruct, bfloat16/4-bit inference; raw prompts $x_1, y_1, z_1 \ \dots \ x_*, y_*, z_*$, with no instruction and no chat template; z candidates are digit tokens $0, \dots, 9$; supports 25, 50, 100 with 256, 256, 512 contexts respectively; x-only, y-only, and intercept-only posterior controls are scored against the same model distributions.
Causal statistic use (Table 8)	Qwen2.5-7B activation patching; layer-specific rank-4 clean-minus-corrupt subspaces; full-vector, learned-subspace, matched-random-subspace, and matched-random-vector interventions; layers 12, 24, 26 swept with layers 24, 26 reported; keybinding $n=256$ evaluation pairs; Bernoulli-next 256 generated and 229 retained after the denominator filter; 2,000 bootstrap resamples over paired examples.
Cross-task (Table 9, Table 14)	MMLU random-permutation audit on Qwen2.5-7B: 1200 items, 8 demonstrations, 16 true random unique demonstration permutations per item, raw per-order probabilities saved, paired item bootstrap CIs and sign-flip p -values. GSM8K-derived multiple choice uses $k \in \{1, 2, 4, 8\}$ random demonstration orderings on Qwen2.5-7B-Instruct.
Evidence QA (Table 15)	Banded permutations with $B=6$ contiguous bands; $m = 12$ unique permutations per item for the 7B model and $m = 16$ for the 8B model; uniform mixture vs. optimized convex weights.

Table 16: **Per-component reproducibility configuration.**